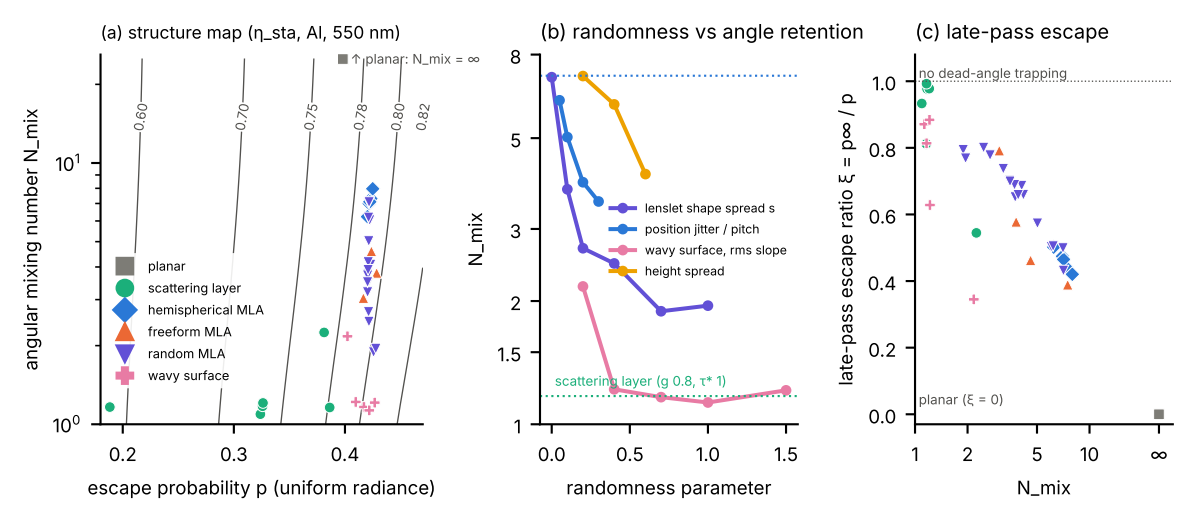
(a) structure map (η_sta, AI, 550 nm)
↑ planar: N_mix = ∞
0.60
0.70
0.75
0.78
0.80
0.82
angular mixing number N_mix
10^1
10^0
planar
scattering layer
hemispherical MLA
freeform MLA
random MLA
wavy surface
0.2
0.3
0.4
escape probability p (uniform radiance)
(b) randomness vs angle retention
N_mix
8
5
3
2
1.5
1
lenslet shape spread s
position jitter / pitch
wavy surface, rms slope
height spread
scattering layer (g 0.8, τ* 1)
0.0
0.5
1.0
1.5
randomness parameter
(c) late-pass escape
no dead-angle trapping
late-pass escape ratio ξ = p∞ / p
1.0
0.8
0.6
0.4
0.2
0.0
planar (ξ = 0)
1
2
5
10
∞
N_mix